

Problem Set #2: Electron diffraction from graphene

Read the paper ‘Grains and grain boundaries in single-layer graphene atomic patchwork quilts’ (Huang, Nature 2011). Notice that the key technique is using electron diffraction to map the graphene crystal lattice in different areas of a polycrystalline sheet.

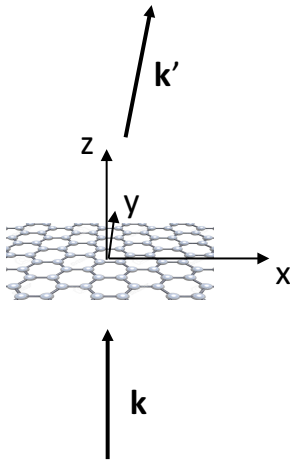
Consider diffraction of 200 keV electrons from a sheet of graphene, at normal incidence as shown to the left.

The incoming electrons have energy $E = \frac{\hbar^2 k^2}{2m} = 200 \text{ keV}$ (m is the mass of the electron). The incoming wavevector is given by $\mathbf{k} = (0, 0, k)$ and the final wavevector is given by $\mathbf{k}' = (k'_x, k'_y, k'_z)$. In a diffraction experiment, the electron energy does not change (only the direction), and the deflection is small. Therefore we can ignore the change in the z wavevector and say that the change in wavevector is given by

$$\Delta \mathbf{k} = \mathbf{k}' - \mathbf{k} \approx (k'_x, k'_y, 0)$$

For a collection of atoms, the direction-dependence of the scattering intensity is given by $I \propto |f|^2$ where $f = \sum e^{i\Delta \mathbf{k} \cdot \mathbf{R}}$, with the sum over the positions $\mathbf{R}$ of the atoms.

- (a) Calculate the magnitude of the wavevector $\mathbf{k}$, in units of inverse Angstroms ($1 \text{ \AA}^{-1} = 10^8 \text{ cm}^{-1}$). A convenient unit conversion is $\frac{\hbar^2}{m_e} = 3.7 \times 10^{-16} \text{ eV cm}^2$.



You will plot the scattering intensity as a 2D color-plot, as a function of k'_x , k'_y , out to $\pm 6 \text{ \AA}^{-1}$. To do this, loop over a grid of values of (k'_x, k'_y) and at each point, sum over all of the atomic positions $\mathbf{R}$ for the graphene sheet. Use matlab, python, or your favorite software to do this.

(b) First consider the fake case where there is one atom at the center of each unit cell. First, try a small number of unit cells, and then try more... how many are needed so that the pattern doesn't change very much when you add more? Make a color plot for this case. Identify the reciprocal lattice vectors corresponding to the main diffraction peaks.

(c) Now consider the real case where there are two atoms per unit cell. Using the same number of unit cells as in (b), make a color plot of the diffraction pattern. Identify the reciprocal lattice vectors. You will notice that not every lattice

vector appears – this is due to destructive interference between the A and B sublattices. Identify which ones are suppressed in this case vs. the case of one atom per unit cell.